ENGR 16100 – FALL 2025

Honors Introduction to Innovation and the Physical Science of Engineering Design I (4 credits)

COURSE INFORMATION:

Sections H01/H02 (CRN 19049/19058): MWF, 9:35 am – 11:20 am; LMBS 5237; "Section 1"
Sections H03/H04 (CRN 19050/19059): MWF, 9:35 am – 11:20 am; Shreve C-111; "Section 2"
Sections H05/H08 (CRN 19055/19061): MWF, 11:35 am – 1:20 pm; Shreve C-111; "Section 3"
Sections H07/H10 (CRN 19056/29459): MWF, 1:35 pm – 3:20 pm; LMBS 5237; "Section 4"
Sections H06/H09 (CRN 19060/29458): MWF, 3:35 pm – 5:20 pm; LMBS 5237; "Section 5"
Instructional Modality: Face-to-Face

Note: Official Purdue sections designations for this class are indicated at the left of each line above. On the right of each line, in quotes, is the informal section designation that will be used by the Instructional Team.

COURSE PREREQUISITES:

Enrollment in this course requires prior admission to both the First-Year Engineering program (or Pre-Agricultural and Biological Engineering program) and the Honors College.

INSTRUCTOR CONTACT INFORMATION:

- Dr. Sean Brophy (sbrophy@purdue.edu): ARMS 1217; 496-3316
- Dr. Dong Hoon Lee (lee1131@purdue.edu): LMBS 2273; 494-7733
- Dr. Timothy Whalen (whalen@purdue.edu): LMBS 2275; 494-4473

Note: all office hours with instructors are by appointment. The best way to set up an appointment is to send an email to your instructor at the address listed above. In your email, please provide 2 or 3 suggestions for blocks of time you would be available for an appointment.

GRADUATE TEACHING ASSISTANT CONTACT INFORMATION:

- Natalie Grambart: ngrambar@purdue.edu
- Abhishek Malhotra: malhot20@purdue.edu
- Grigorii Rudakov: grudakov@purdue.edu
- Cody Waldecker: cwaldeck@purdue.edu
- Sam Yang: smyang@purdue.edu

COURSE OFFICE HOURS:

Mondays, 5:30-7:30 pm (online)
Tuesdays, 6:30-8:30 pm (in-person @ SHRV C-111)
Wednesdays, 5:30-8:30 pm (in-person @ SHRV C-111)
Thursdays, 4:30-8:30 pm (in-person @ SHRV C-111)

No office hours will be held during the first week of classes.

COURSE DESCRIPTION & LEARNING OBJECTIVES:

This course introduces students to the engineering profession using physics-based, multidisciplinary, societally relevant content. Students develop engineering approaches to systems, generate and explore creative and innovative ideas, and use computational methods to support design decisions. In particular, the students will develop the ability to model and investigate physical systems with a focus on vector analysis, linear momentum, angular momentum, work-energy, and solid material interactions. Design challenges and projects will explore a wide range of natural phenomena experimentally and computationally (utilizing Matlab and Python) and engage students in innovative thinking across the engineering disciplines at Purdue. They will learn the basics of descriptive statistics, data analysis, sensitivity analysis, and decision making. Students experience the process of design and analysis in engineering including how to work effectively in teams. Students also develop skills in project management, engineering fundamentals, oral and graphical communication, logical thinking, and modern engineering tools. Typically offered Fall.

Successful completion of this course as well as ENGR 16200 (typically offered in the Spring semester) will meet the Foundational Learning Outcome of Information Literacy for Purdue University's Core Curriculum. Also, successful completion of this course as well as ENGR 16200 counts as one course towards meeting the Foundational Learning Outcome of Science for the Core Curriculum.

Successful completion of this course will enable a student to:

1. Discuss the engineering education process, courses, and options, ask informed questions about engineering job functions and opportunities, and use this information to prepare a moderately well-informed course of study for academic and career success.
2. Employ formal strategies, including project organization and time management techniques, to thoughtfully pursue course activities.
3. Plan and implement systematic design processes using engineering management and design tools such as work breakdown structures, technical requirements and functional block diagrams to design innovative products and systems.
4. Perform an appropriate process for defining an engineering design problem, including identifying design factors in need of further information, using relevant information sources to research these factors, and defining quantitative metrics that can be used to determine successful fulfillment of a design factor.
5. Create analytical and/or computational models of engineering systems that can be employed to better understand or control the systems.
6. Describe a wide range of engineering and physical phenomena using fundamental physical laws.
7. Translate logical procedures or processes into algorithms that can be implemented in computational environments.

8. Demonstrate professional communication skills in the areas of technical writing to formulate responsive engineering reports with appropriate scope and attribution of the original work and ideas of others via proper methods, in presentations to convey relevant engineering evidence and information verbally and graphically to audiences, and in record-keeping to document the design process for other engineers.
9. Apply knowledge regarding effective team roles, norms, and stages of development to work alongside individuals with diverse backgrounds and accomplish engineering tasks.
10. Evaluate engineering problems to obtain and communicate evidence-based conclusions by synthesizing one or more sources of information and utilizing data interpretation skills to identify credible results.
11. Discuss how social, cultural, safety, and sustainability issues affect the engineering design process, including impacts on resources, the need for appropriate information, and the necessity to question default assumptions.
12. Incorporate the consideration of engineering ethics into instances of engineering thinking, problem solving, and communication so that the broader impacts of engineering work are evaluated and accounted for and that information is expressed in ways that follow proper practices for using confidential information and the proprietary ideas of others.
13. Apply fundamental engineering skills and knowledge relating to units, dimensions, estimation, significant digits, vectors, and the problem presentation method to engineering applications.
14. Display proficiency in the applications of engineering content knowledge including computer-aided drawing and basic statistics.

COURSE EXPECTATIONS:

You are expected to:

1. Be an active contributor to in-class discussion and problem-solving activities.
2. Be prepared and accountable for class by reading any assigned material ahead of time and be able to answer questions related to this material.
3. Be held accountable for all material in reading assignments, course documents, electronic communications, and other sources of course information, regardless of whether the material is explicitly discussed in class or not.
4. Attend class because of a community expectation to learn from each other.
5. Be cooperative with your team and work with them, not compete against them.
6. Learn interdependently with your team and your peers.
7. Learn to be accountable to your team and have your team be accountable to you.
8. Be prepared to meet with your team outside of class to complete assignments.
9. Rely on your peers, as well as the instructional team, to learn the course material.

COURSE GRADE COMPUTATION:

3 Midterm Exams; each worth 12%	36%
Homework	22%
Ready Assessment Trials (RAT's)	3%
Checks for Understanding (CFU's)	7%
Design Challenges and Lab Activities	4%
Project 1	3%
Project 2	8%
Project 3	17%
Total	100%

See below, along with the sections "Engineering Your Career Assignment" and "Attendance", for additional information regarding how your course grade can be affected.

COURSE GRADE COMPUTATION NOTES:

This course will make extensive use of student teams. As such, homework, activities, ready assessment trials (RAT's), checks for understanding (CFU's), and project grades may reflect some combination, in part or as a whole, of your individual effort and teamwork. Exam grades will, in their entirety, represent your individual understanding of the course material. In general, your final course grade will consist of no less than 68% of your own individual contributions. You are reminded that learning team accountability (your accountability to the team and the team's accountability to you) is an essential element of this course. As such, the course instructor reserves the right to use: materials submitted by your team to reflect your individual effort in the form of a grade; materials submitted by individuals to reflect your team's effort in the form of a grade; materials randomly collected by individuals to reflect your team's effort in the form of a grade; the weakest material submitted by individuals to reflect your team's effort in the form of a grade; or materials submitted by pairs of team members to reflect your individual or team effort in the form of a grade.

The following grading scale will be used to determine your semester course grade:

$$90\% \leq A < 100\%,\ 80\% \leq B < 90\%,\ 70\% \leq C < 80\%,\ 60\% \leq D < 70\%, \text{ and } F < 60\%$$

You are, at a minimum, guaranteed a letter grade for the course that corresponds to your course grade percentage as determined by the course grade computation (see above). However, the course instructor reserves the right to review a borderline student on a case-by-case basis. Factors that may be used to evaluate such cases may include: class attendance, participation, teaming, grade improvement, consistent homework completion, bonus completion, and the like.

Consistent with the policies described below, failure to complete the "Engineering Your Career" (EYC) assignment will result in a deduction of one letter grade, regardless of your current numeric score. Also, having an excessive number of unexcused absences will result in a deduction of one letter grade, regardless of your current numeric score.

REQUIRED TEXT BOOK:

There is no required textbook to be purchased for this course. Reading assignments will be made available online via Brightspace or distributed to students in class during the course.

OTHER REFERENCE MATERIALS:

These materials are recommended references:

1. National Academy of Engineering, "Grand Challenges for Engineering", available at http://www.engineeringchallenges.org/, last accessed 08/2016.
2. Programming references for Fall:
   - Programming in Python 3, 2nd Edition
     Author: Summerfield
     Publisher: Addison Wesley
     ISBN 978-0-321-68056-3

IMPORTANT DATES: (EXAM, CFU, AND PROJECT DATES ARE TENTATIVE)

August 29 – Project 1 released

September 8 – Labor Day (no classes)
September 12 – CFU #1 (in class)
September 17 – Project 1 Demonstration (outside of class)
September 19 – Project 1 Executive Summary to be submitted electronically
September 22 – Project 3 released
September 29 – Exam #1 (in class)

October 13 – October Break (no classes)
October 22 – CFU #2 (in class)
October 29 – Exam #2 (in class)
October 31 – Project 2 released

November 10 – Project 2 Presentations (in class)
November 12 – Project 2 Report to be submitted electronically
November 17 – CFU #3 (in class)
November 26 - 28 – Thanksgiving Break (no classes)

December 3 – Exam #3 (in class)
December 8 – Project 3 Demonstration (outside of class)
December 10 – Project 3 Presentations (in class)
December 12 – Project 3 Report to be submitted electronically

ENGINEERING YOUR Career (EYC) Assignment:

The Engineering Your Career (EYC) assignment is an opportunity for you to learn more about the various options and potential careers in selected fields of engineering, as well as having you investigate how Purdue's engineering degree programs prepare you for these fields. You will be provided with specific details concerning this assignment during the first week of class, including requirements, due dates, and other relevant details. Because of the potential impact that this assignment can have on your academic and professional career, it is **mandatory** to complete this assignment in a satisfactory manner. Students who do not complete this assignment in a satisfactory manner will be given a one letter grade deduction for the class **regardless** of their performance on other aspects of the course.

ATTENDANCE:

As indicated in the Course Expectations, you are expected to attend ALL classes. Students should contact their instructor (not a Grad TA or Peer TA) to make the instructor aware of any situation in which the student is experiencing a significant illness (or similar health-related event) and discuss further steps as needed.

In keeping with Purdue's Class Attendance and Absence Reporting Policy (see https://www.purdue.edu/advocacy/students/absences.html), only your instructor can excuse you from classes or course responsibilities; no one else has this authority. For this course, excusable absences are limited to: medically excused absences; extended absences as documented by the Office of the Dean of Students; or Purdue sponsored events (e.g., conferences, club activities, varsity athletics, etc.) that do not conflict with an exam. Other situations (e.g., interviews, religious observances, personal events, etc.) may be excused at the discretion of your instructor. Be aware that having grounds for an excusable absence does not necessarily guarantee that you will obtain instructor permission to miss class or course responsibilities.

In the case of foreseeable events (e.g., Purdue sanctioned events with known dates, scheduled medical procedures, etc.), students are expected to provide appropriate documentation of the event prior to its occurrence with sufficient time for the instructor to make a decision about the suitability of the absence. In particular, if the event is Purdue sanctioned, students should have a person in authority with respect to this event send a request to your instructor to excuse the absence. In general, the person in authority should be a member of the faculty, staff, or administration; however, requests from other persons will be considered on a case-by-case basis. Instructors reserve the right to decline to excuse an absence, with the understanding that such a decision must be consistent with the Purdue policy referenced above.

In all cases where a student is requesting an excused absence, the student is expected to make a good faith effort to notify the course instructor in a timely fashion (prior to a class is considered timely, after class is generally not timely unless there are extenuating

circumstances). The instructor may request further documentation to corroborate your request. Once an absence is approved, you will have one week to make arrangements with your course instructor for a means of making up any material you missed.

When the student is unable to make direct contact with the instructor and is unable to leave word with the instructor because of circumstances beyond the student's control, and in cases of bereavement, quarantine, or isolation, the student or a representative should contact the Office of the Dean of Students via email at odos@purdue.edu or phone at 765-494-1747.

Please note: students having an excessive number of unexcused absences (defined as five or more of the regularly scheduled class meetings) will automatically be dropped one letter grade regardless of their class standing.

ACADEMIC INTEGRITY:

Academic integrity is one of the highest values that Purdue University holds (See https://www.purdue.edu/odos/osrr/academic-integrity/index.html). Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

Purdue University maintains several policies related to academic integrity. For your convenience, you can find a link to Purdue's Student Guide for Academic Integrity on the course Brightspace page at *Content > Start Here > University Policies and Statements*.

Please pay special attention to the following paragraph regarding teamwork (or working collaboratively with others). During this course, you will be working in teams and as such you are expected, and will be encouraged, to help each other. This is done because it has been shown that students learn more effectively while working together. Since course grades are not curved, there is no penalty for helping someone else. However, there is, at times, confusion over when it is OK to "collaborate with a teammate (or someone in the course)" and when collaborating with someone else turns into academic dishonesty.

When an assignment specifies that it is:

1. **ALL-CLASS Assignment** – you should feel comfortable talking to anyone in the course (and working side-by-side with them) about any aspect of an assignment from gaining conceptual insight to developing an appropriate model to specifying assumptions to writing out a solution. If the assignment was to develop some kind of computer tool model/solution, working side-by-side with other members of the course to gain conceptual insight, develop logic, outline syntax, and implement/debug said logic and syntax would be considered acceptable behavior. In

such cases all individuals involved in the assignment should be appropriately acknowledged in the materials submitted.

2. **TEAM Assignment** – you should feel comfortable talking to anyone on your team (and working side-by-side with them) about any aspect of an assignment from gaining conceptual insight to developing an appropriate model to specifying assumptions to writing out a solution. If the assignment was to develop some kind of computer tool model/solution, working side-by-side with other members of your team to gain conceptual insight, develop logic, outline syntax, and implement/debug said logic and syntax would be considered acceptable behavior. In such cases all individuals involved in the assignment should be appropriately acknowledged in the materials submitted.
3. **INDIVIDUAL Assignment** – you should feel comfortable talking to anyone in the course about an assignment <u>to gain conceptual insight only</u>. Any act other than having a "conceptual conversation," even if mutually agreed upon, would be considered academic dishonesty. If the assignment was to develop some kind of computer tool model/solution, working with others to gain conceptual insight would be considered acceptable behavior. Any act other than having a "conceptual conversation" or "providing debugging insight," even if mutually agreed upon, would be considered academic dishonesty.

POLICIES REGARDING USE OF ARTIFICIAL INTELLIGENCE/LARGE LANGUAGE MODELS AND ACADEMIC INTEGRITY

As you are likely already aware, developments in the field of artificial intelligence and large language models have led to the creation of powerful tools (*e.g.,* ChatGPT) for the creation of content. This includes diverse areas such as generating reports, writing computer programs, solving homework problems, etc. A natural question for students is to what extent they can take advantage of such tools in this class.

Our general philosophy is that tools such as ChatGPT should not be expressly prohibited from use in this class, as it is likely that future engineers will need to develop strong skills in using AI/LLMs in professional situations. However, it is also vital that students actually learn the course material for themselves and should not become overly reliant on these tools as a substitute for developing an understanding of concepts.

Based on these ideas, the following policies should be followed in this class when using an AI/LLM tool:

- When working on an assignment, the policies regarding All-Class, Team, and Individual assignments should be followed by treating an AI/LLM tool in the same way one would treat a teammate or another student in the class. For example, when working on an Individual assignment, ChatGPT can be consulted to gain conceptual insight into a

problem but should not be used to develop a model, implement/debug the logic of a computer program, write out a solution, etc.

- AI/LLM tools will not be available for use in exams, CFU's, or RAT's. Students are responsible for creating their own understanding of the material relevant to the exam, CFU, or RAT and being able to demonstrate this understanding appropriately.

OTHER EXPECTATIONS, RULES, OR COMMENTS:

1. A "Failure to Follow Instructions" policy is in place, which penalizes you for repetitive or flagrant violations of assignment instructions. After two repetitive or flagrant violations of assignment instructions, your assignment will automatically be returned with a grade of zero (0) with no chance of redress.
2. Submission of course assignments is subject to the policies in the table below. All submissions must be made using the mechanism indicated in the assignment or course materials; submissions made via a different mechanism can be rejected at the discretion of the course instructor. In certain circumstances an instructor can waive a given submission policy or mechanism **prior** to the submission; students are expected to contact the appropriate person immediately if they wish to request an exception to a certain policy.
3. When submitting course assignments electronically, only the final submission will be graded. You should include all relevant files in the final submission. It is recommended that you confirm all electronic submissions have been successful to ensure that your work will be graded.
4. Hand-written work:
   - All hand-written homework will be submitted on engineering paper.
   - All hand-written work will be submitted with your name, your team number and section number printed in the upper right-hand corner of your paper. In addition, you must sign your work below your name. Your signature indicates "*this is my work, and I have a general understanding of all the information being submitted*."

Deliverable Type	Policies
Exams	• Penalties for late submission at instructor's discretion. • Missed exams due to an excused absence must be made up within one week from original exam date unless otherwise indicated by instructor. • Students **must** contact their instructor as soon as possible after the exam to determine steps for making up an exam.
Readiness Assessment Tests (RATs)	• No late submissions accepted. • Missed RATs cannot be made up.

Homework Assignments Design Challenges	• Flat 30% penalty if submitted within 24 hours of on time. • No submissions accepted after 24 hours of on time.
Check For Understanding (CFU's)	• Penalties for late submission at instructor's discretion. • Missed CFUs due to an excused absence must be made up within one week from original CFU date unless otherwise indicated by instructor. • Students **must** contact their instructor as soon as possible after the CFU to determine steps for making up the CFU.
Project Reports Project Presentations	• 5% of deliverable value for up to 15 minutes after the time due, **assessed at timestamp of GTA email receipt**. • After 15 minutes, flat 30% penalty for submission up to 24 hours late. • No submissions accepted after 24 hours.
Bonus Assignments Project Notebooks Project Design Spec Reviews Requests For Additional Information (RFAI)	No late submissions accepted.
Kit inventory sheets PoC specifications Other not-for-credit Project deliverables	1% **OVERALL** project score penalty

- When submitting a team hand-written homework, you should follow the same rules as stated above, except making sure to include the names of all the team members that participated. In the case of a team assignment, the signature of each individual below his/her name communicates "*I was an active participant in preparing this document and I have a general understanding of all the information that is being submitted*."

5. Computer Assignments:
   - You need to follow the Code Standard associated with the particular computer tool to receive maximum credit.
   - Computer assignments will always be submitted using the appropriate header that includes your name, your team number, and your section ID.
   - You will always provide an electronic signature (signature: your full name). Your electronic signature on an individual homework assignment indicates, "*this is my work, and I have not collaborated with other individuals (other than the teaching team) to obtain the final materials being submitted."* Your electronic signature

on a team homework assignment indicates, "*I was an active participant in preparing the materials and I have a general understanding of all the information that is being submitted*."

6. A late penalty will be assessed to any team assignment submitted late because a team member fails to act responsibly, even if it was completed on time.

STUDENT SUPPORT AND RESOURCES:

The Brightspace page for this course contains several important links to student support services and related information. These can be found at *Content > Start Here > Student Support and Resources.* Please review these links to become aware of the wide variety of Purdue resources available to you to assist in many different academic situations.

Specific information found here includes the following:

- How to access the Student Services Widget
- Mental Health, Wellness and Basic Needs Security statement
- Strategies and tips for Engaging in your Learning
- Purdue's Web Accessibility Policy and Accessibility Standards in Brightspace

UNIVERSITY POLICIES AND STATEMENTS:

The Brightspace page for this course also contains several important links to statements of official Purdue University policies. These can be found at *Content > Start Here > University Policies and Statements.* Please review these links to familiarize yourself with your rights and responsibilities as a Purdue student for a range of academic and personal scenarios.

Specific information found here includes the following:

- Purdue's Student Guide for Academic Integrity (previously mentioned in the Academic Integrity section)
- Nondiscrimination Policy Statement
- Office of the Dean of Students Class Absence policies and Academic regulations concerning attendance
- Use of Copyrighted Materials statement
    - Please note that almost all documents produced by the Instructional Staff are copyrighted and thus are subject to Purdue University's policies on the use of copyrighted materials.
    - All other works produced by the Instructional Staff are considered to be derivative works and thus subject to the policies outlined in Part 10 of the Purdue Student Miscellaneous Conduct regulations.
    - In addition, notes taken by students in class are considered to be derivative works and subject to the same policies.
- Information on Emergency Preparedness

    - In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email.
- Violent Behavior Policy
- Amorous Relationships policy